Bayesian adaptive distributed lag models

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Abstract

Distributed lag models (DLMs) express the cumulative and delayed dependence between pairs of time-indexed response and explanatory variables. In practical application, users of DLMs examine the estimated influence of a series of lagged covariates to assess patterns of dependence. Much recent methodological work has sought to develop flexible parameterisations for smoothing the associated lag parameters that avoid overfitting. However, this paper finds that some widely-used DLMs introduce bias in the estimated lag influence, and are sensitive to the maximum lag which is typically chosen in advance of model fitting. Simulations show that bias and misspecification are dramatically reduced by generalising the smoothing model to allow varying penalisation of the lag influence estimates. The resulting model is shown to have substantially fewer effective parameters and lower bias, providing the user with confidence that the estimates are robust to prior model choice.

Keywords: Distributed lag model; penalized spline; adaptive smoothing

1 Introduction

Distributed lag models (DLMs) find application whenever the influence of a time-indexed independent variable is delayed and spread over time. Although DLMs are widely applicable, their development was originally motivated by problems in econometrics [Nerlove, 1958, Almon, 1965, Zellner and Geisel, 1970, Haugh and Box, 1977] and they have more recently experienced a surge in popularity in environmental epidemiology [Zanobetti et al., 2000, Welty et al., 2009, Gasparrini et al., 2010]. Crucially, DLMs enable direct interpretation of the influence of a temporal exposure ensemble, which is particularly useful for characterising the total health impact of persistent environmental exposures such as air pollution or temperature [Wyzga, 1978, Gasparrini et al., 2010]. Lag models can help to identify subtle types of time-dependence such as 'mortality displacement', which occurs when exposure related mortality diminishes a vulnerable subpopulation, resulting in lower mortality in subsequent time intervals. Mortality displacement has been widely documented [Schwartz, 2000, Braga et al., 2001] and is conspicuous where lag influence is estimated to have a protective effect at low or moderate lags for what are otherwise harmful exposures [Zanobetti

et al., 2000]. Correctly identifying these effects depends on obtaining unbiased estimates of the underlying lag influence, and it is therefore essential that the DLM is correctly specified. Since no previous study has formally quantified the estimation cost of model misspecification, this paper provides a comparison of various existing techniques.

If a time-varying explanatory variable X_t has some influence on a response variable Y_t over a set of lags $0, \ldots, p$, then a simple DLM can be expressed as the regression model $Y_t = \sum_{i=0}^{p} X_{t-i}\beta_i + \epsilon$, where ϵ is an independent error process and the parameters $\boldsymbol{\beta} = (\beta_0, \beta_1, \ldots, \beta_p)$ are referred to as the 'lag curve' or lag function, as a function of i. $\boldsymbol{\beta}$ cannot be reliably estimated using ordinary generalised linear models when X_t and X_{t-1} are correlated, and regularisation using shape constraints or smoothness restrictions over the lag parameters $\boldsymbol{\beta}$ are frequently used. Recent studies have implemented various forms of flexible parameterisation using polynomial splines and smoothness penalties, for example by combining uniformly spaced B-splines with ridge penalisation [Zanobetti et al., 2000], logarithmically spaced B-splines [Gasparrini et al., 2010], penalised B-splines [Rushworth et al., 2013, Obermeier et al., 2015], and Bayesian penalised splines with varying ridge penalties [Muggeo, 2008]. Spline basis representations ensure a minimum level of smoothness, and offer computational relief when there are fewer knots than lags, which reduces the total number of parameters to estimate. Although smoothness restrictions primarily serve to stabilise the model, they can also be exploited to incorporate additional structure over the shape of the estimated lag curve. For example, it is widely assumed that the true lag curve has more curvature at short lags than at large lags, reflecting the intuition that recent events are generally the most influential, and influence should decay rapidly to zero for increasing lags. Further structure can also be used to avoid discontinuity between the influence of the final modelled lag $\beta_p \neq 0$ and the implicitly assumed $\beta_{p+j} = 0$. Some examples include Koyck [1954] who restricted $\boldsymbol{\beta}$ to decay geometrically with increasing p; Muggeo [2008] who combined a B-spline basis with parameters which were shrunk by a lag-varying ridge penalty; Gasparrini et al. [2010] used logarithmic knot placement in the lag dimension to enable varying smoothness; and Welty et al. [2009] constructed a Gaussian process prior distribution that enforces smaller covariances on lag parameters at short lags than longer ones. While each of these approaches achieves a level of lag-varying smoothness, each incorporates a parametric assumption on the type of penalisation that occurs, which may not be appropriate in general.

A further problem arises because the maximum lag p must be chosen before model fitting: this is sometimes justified by drawing on previously published studies, for example, in several studies of the delayed effects of temperature on human health, $20 \leq p \leq 30$ is consistently reported [Braga et al., 2002, Armstrong, 2006, Gasparrini and Armstrong, 2013]. In general, there is a cost to getting p wrong: too large and the model risks overfitting, and too small can bias the resulting lag function estimate. Few studies have investigated sensitivity of different models to choices of p, although several authors recognise that this is an ongoing difficulty for DLMs. Heaton and Peng [2012] and Heaton and Peng [2014] provide notable exceptions, and estimate p as part of a two-stage approach. However, DLMs are often applied in settings where the true lag curve is believed to decay gradually to 0, and in such settings, p is likely to be poorly estimated and will lack a meaningful interpretation.

This paper proposes a pragmatic strategy that ensures flexible and automatic smoothing over the lag structure that avoids the need to either estimate or choose p in advance. By exploiting the property that very large lags have close to 0 influence, an automatic lag-dependent smoother is constructed that ensures strong penalisation at high lags so that p can be made arbitrarily large without the risk of overfitting. This follows the tradition of richly specifying a hierarchical model and ensuring parsimony by liberal use of regularisation and smoothing. A new DLM is described in Section 2 and in Section 3 a comprehensive simulation study illustrates the DLM's desirable properties, especially in comparison to existing approaches. This study compares bias resulting from getting p wrong, which is itself a central contribution, as DLMs have not previously been compared in this manner. Section 4 applies the model to four real data sets: daily stock market returns, intraday order flow and prices at one-second resolution, air pollution and mortality in Chicago, and COVID-19 cases and deaths. Section 5 concludes the paper with an overview of the main findings and a discussion of the implications for users of distributed lag models.

2 Adaptive distributed lag models

Define $\mathbf{y} = (y_1, \ldots, y_n)$, a response variable time series of length n, and $\mathbf{x} = (x_1, \ldots, x_n)$, a time series of an exposure or input, whose influence on $\mathbf{y}$ up to some lag p is to be estimated. A Gaussian DLM for these data can be written as $y_i \sim \mathrm{N}\left(\sum_{j=0}^{p} x_{i-j}\beta_j,\ \sigma^2\right)$ where $i = p+1, p+2, \ldots, n$,

and σ^2 is the error variance. The lag parameters $(\beta_0, \beta_1, \ldots, \beta_p)$ could be penalised and estimated directly, but for ease of comparison with existing approaches, these will be projected onto a B-spline basis of K functions uniformly spaced over the lag range $0, \ldots, p$. The basis representation can be expressed as $\beta_j = \sum_{k=1}^{K} B_k(j) b_k$ where the further set of parameters $\mathbf{b} = (b_1, \ldots, b_K)$ determine the fitted lag curve. The model can be expressed as $\mathbf{y} \sim \mathrm{N}(\mathbf{X}\mathbf{b},\ \sigma^2\mathbf{I})$, where $\mathbf{X}_{ik} = \sum_{j=0}^{p} x_{i-j} B_k(j)$. To control the roughness of b_k in a way that enables variable smoothing, a similar approach to Brewer and Nolan [2007] and Reich and Hodges [2008] is used, in which the intrinsic autoregressive (ICAR) prior distribution of Besag et al. [1991] is generalised. The conditional prior distribution for $\mathbf{b}$ under the ordinary ICAR is $\mathbf{b} \sim \mathrm{N}(\mathbf{0}, \frac{1}{\lambda}\mathbf{P}^{-})$, resulting in a conditional prior density of

$$\pi(\mathbf{b} \mid \lambda) \propto \exp\left(-\frac{\lambda}{2}\mathbf{b}^{\top}\mathbf{P}\mathbf{b}\right) = \exp\left(-\frac{1}{2}\sum_{k=1}^{K-1}\lambda(b_{k+1} - b_k)^2\right). \tag{1}$$

The precision parameter λ controls the extent to which the squared differences of consecutive pairs of $\mathbf{b}$ influence the posterior, imparting a smoothing effect over the lag curve derived from $\mathbf{b}$. Since each squared difference $(b_{k+1} - b_k)^2$ contributes equally in the prior in Equation (1), variable smoothing can be achieved naturally by introducing a set of precision parameters $\boldsymbol{\lambda} = (\lambda_1, \ldots, \lambda_{K-1})$ that enables pairs of B-spline parameters to be smoothed more or less strongly. This generalised prior can be written $\pi(\mathbf{b} \mid \boldsymbol{\lambda}) \propto \exp\left(-\frac{1}{2}\sum_{k=1}^{K-1}\lambda_k(b_{k+1} - b_k)^2\right) = \mathrm{N}(\mathbf{0}, \mathbf{Q}^{-})$ where $\mathbf{Q} = \mathbf{D}^{\top}\mathrm{diag}(\boldsymbol{\lambda})\mathbf{D}$, with $\mathbf{D}$ the $(K-1) \times K$ first-difference matrix, is the precision matrix

$$\mathbf{Q} = \begin{pmatrix} \lambda_1 & -\lambda_1 & 0 & 0 & \ldots & 0 \\ -\lambda_1 & \lambda_1 + \lambda_2 & -\lambda_2 & 0 & \ldots & 0 \\ 0 & -\lambda_2 & \lambda_2 + \lambda_3 & -\lambda_3 & \ldots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & -\lambda_{K-1} & \lambda_{K-1} \end{pmatrix}.$$

$\mathbf{Q}$ has rank $K - 1$ and its null space is spanned by the constant vector, so the prior is intrinsic: it is improper with respect to the overall level of $\mathbf{b}$ but proper for the differences, and $\mathbf{Q}^{-}$ denotes a generalised inverse. The posterior distribution is nevertheless proper, since the level of $\mathbf{b}$ is

identified by the likelihood. This joint specification for $\mathbf{b}$ results in the conditional distributions

$$b_k \mid \mathbf{b}_{-k}, \boldsymbol{\lambda} \sim \begin{cases} \mathrm{N}\left(b_{k+1},\ \frac{1}{\lambda_1}\right) & \text{if } k = 1 \\ \mathrm{N}\left(\frac{\lambda_{k-1} b_{k-1} + \lambda_k b_{k+1}}{\lambda_{k-1} + \lambda_k},\ \frac{1}{\lambda_{k-1} + \lambda_k}\right) & \text{if } k = 2, \ldots, K-1 \\ \mathrm{N}\left(b_{k-1},\ \frac{1}{\lambda_{K-1}}\right) & \text{if } k = K. \end{cases}$$

When $\lambda_1 = \lambda_2 = \ldots = \lambda_{K-1} = \lambda$, then $\mathbf{Q} = \lambda \mathbf{P}$, which shows that the ordinary ICAR prior in Equation (1) is a special case of the adaptive prior.

The influence of very large lags is expected to be negligible, and a lag curve that terminates away from zero at lag p would imply an artificial discontinuity with the implicitly assumed $\beta_{p+j} = 0$. Rather than modifying $\mathbf{Q}$, this is discouraged by a single pseudo-observation: the row $(B_1(p), \ldots, B_K(p))$ is appended to $\mathbf{X}$ together with a zero response, which contributes the factor $\exp\{-\beta_p^2/(2\sigma^2)\}$ to the posterior and so shrinks the final value of the lag curve towards zero at the scale of the error variance, without introducing an additional tuning parameter. Alternatively the constraint $\beta_p = 0$, optionally together with $\beta_p' = 0$, can be imposed exactly by restricting $\mathbf{b}$ to the null space of the constraint rows, which leaves the generalised determinant above unchanged up to a constant; this is the default in the accompanying software.

A further smoothing prior distribution is placed over $\boldsymbol{\lambda}$ to restrict their flexibility. This is particularly attractive as it also reflects our intuition that the curvature of the lag function should not change too rapidly. For this purpose, an ordinary ICAR prior is used to smooth the log precisions $\boldsymbol{\tau} = (\tau_1, \ldots, \tau_{K-1})$, $\tau_k = \log \lambda_k$, such that $\boldsymbol{\tau} \mid \zeta^2 \sim \mathrm{N}(\mathbf{0}, \zeta^2 \mathbf{K}^-)$ where $\mathbf{K}$ is the $(K-1) \times (K-1)$ precision matrix

$$\mathbf{K} = \begin{pmatrix} 1 & -1 & 0 & 0 & \ldots & 0 \\ -1 & 2 & -1 & 0 & \ldots & 0 \\ 0 & -1 & 2 & -1 & \ldots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & \ldots & 1 \end{pmatrix}.$$

Therefore, for small values of the variance ζ^2, pairs of consecutive elements of $\boldsymbol{\tau}$ are more strongly smoothed towards each other. Finally, the variance parameters were assigned weakly informative

inverse-gamma priors, $\sigma^2 \sim \mathrm{IG}(0.001, 0.001)$ and $\zeta^2 \sim \mathrm{IG}(1, 1/2)$, the values used in the accompanying R package. The combination of adaptive smoothing with smoothed variances as described results in a very flexible model that can accommodate complex shapes using a relatively simple specification. The model described above assumes Gaussian errors for simplicity, but the ideas easily generalise to accommodate other response distributions.

Inference for the models fitted in the simulation study in Section 3 was achieved using Markov chain Monte Carlo (MCMC) simulation using a mixture of Gibbs sampling and Metropolis–Hastings steps, implemented in R [R Core Team, 2016]. The accompanying software, the Python package `badlm` (`https://github.com/alastairrushworth/badlm`), uses a different scheme that is considerably more efficient. Given $\boldsymbol{\tau}$ and σ^2 the model is linear and Gaussian, so $\mathbf{b}$ can be integrated out analytically: with $\mathbf{A} = \mathbf{Q} + \mathbf{X}^\top\mathbf{X}/\sigma^2$ and $\mathbf{m} = \mathbf{X}^\top\mathbf{y}/\sigma^2$,

$$\log p(\mathbf{y} \mid \boldsymbol{\tau}, \sigma^2) = -\frac{n}{2}\log(2\pi\sigma^2) + \frac{1}{2}\log|\mathbf{Q}|_+ - \frac{1}{2}\log|\mathbf{A}| + \frac{1}{2}\mathbf{m}^\top\mathbf{A}^{-1}\mathbf{m} - \frac{\mathbf{y}^\top\mathbf{y}}{2\sigma^2},$$

where $|\mathbf{Q}|_+$ is the generalised determinant of the intrinsic prior precision, the product of its non-zero eigenvalues. Since the pairs (b_k, b_{k+1}) penalised by $\mathbf{Q}$ form a chain, the matrix-tree theorem gives $|\mathbf{Q}|_+ = K\prod_{k=1}^{K-1}\lambda_k$, so this term is simply $\log K + \sum_k \tau_k$. The resulting marginal posterior of $(\boldsymbol{\tau}, \sigma^2)$ is smooth and low dimensional, and is sampled with the No-U-Turn Sampler [Hoffman and Gelman, 2014] using automatic differentiation; $\mathbf{b}$ is then drawn from its Gaussian conditional distribution for each retained draw. Collapsing removes the strong dependence between $\mathbf{b}$ and $\boldsymbol{\tau}$ that slows a Gibbs sampler, and each likelihood evaluation costs $O(K^3)$ regardless of n because only the sufficient statistics $\mathbf{X}^\top\mathbf{X}$ and $\mathbf{X}^\top\mathbf{y}$ are needed. The software also provides a Gibbs sampler with slice updates of $\boldsymbol{\tau}$ as an independent check, a Laplace approximation to the collapsed posterior that serves as a fast approximate fit and as the starting point of the chains, an exact version of the end constraint, AR(1) errors and Poisson and negative binomial responses.

3 Simulation study

The adaptive model detailed in Section 2 was compared with several currently available distributed lag models, labelled $\mathcal{M}_1$, $\mathcal{M}_2$, $\mathcal{M}_3$, $\mathcal{M}_4$ and $\mathcal{M}_5$. $\mathcal{M}_1$ is a non-adaptive version of the DLM

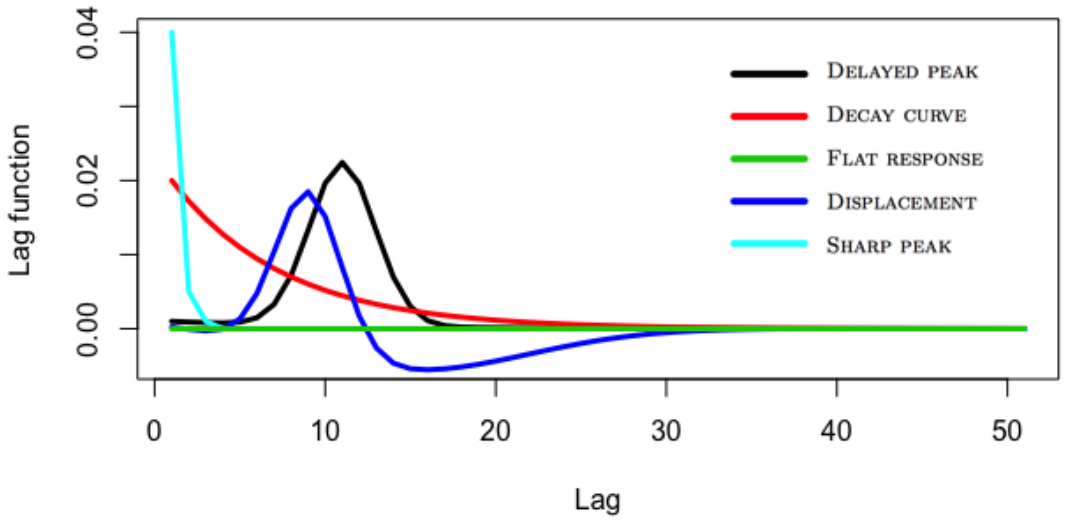


Figure 1: The five lag curves used to simulate response and covariate data with different lag response.

described in Section 2 (achieved by fixing $\lambda_1 = \lambda_2 = \ldots = \lambda_{K-1} = \lambda$), $\mathcal{M}_2$ is a Bayesian implementation of the DLM with linearly increasing ridge penalty proposed by Muggeo [2008], $\mathcal{M}_3$ is the new adaptive model described in Section 2, $\mathcal{M}_4$ is the P-spline model of Obermeier et al. [2015] and $\mathcal{M}_5$ is the DLM with logarithmically spaced knots of Gasparrini et al. [2010].

3.1 Comparison under different lag structures

To compare estimation performance of models $\mathcal{M}_1$–$\mathcal{M}_5$, time series response and covariate data were simulated assuming each of five different-shaped lag relationships, each with maximum lag $p = 50$. The curves are shown in Figure 1 and included a delayed peak scenario (Delayed peak), an exponential decay curve (Decay curve), a horizontal line at 0 equivalent to a null effect (Flat response), a curve exhibiting a shape similar to a mortality displacement effect (Displacement) and a sharply peaked function that declines rapidly to 0 (Sharp peak).

A covariate time series, x_t, was generated by assuming $x_t \sim \mathrm{N}(0.5x_{t-1}, 0.1^2)$, and conditional on each lag function, a response series y_t was generated using $y_t = \sum_{j=0}^{50} \beta_j x_{t-j} + \epsilon_t$ for $t = 51, \ldots, 500$. The error term ϵ_t assumed autocorrelation such that $\epsilon_t \sim \mathrm{N}(0.2\epsilon_{t-1}, 0.1^2)$, whose parameters were selected for an acceptable signal-to-noise ratio. The autocorrelation in both x_t and ϵ_t reflects the temporal dependence in covariates and errors typically observed in real data.

Models $\mathcal{M}_1$–$\mathcal{M}_5$ were fitted to each of 200 realisations of the processes $\{x_t, y_t\}$ described above.

Lag curve shape	Summary	$\mathcal{M}_1$	$\mathcal{M}_2$	$\mathcal{M}_3$	$\mathcal{M}_4$	$\mathcal{M}_5$
	RMSE	17.31	14.53	10.32	15.51	22.68
Delayed peak	Bias^2	3.31	3.24	3.32	9.59	3.83
	ED	18.53	14.99	7.12	12.14	15.81
	RMSE	16.57	11.59	8.45	9.46	14.15
Decay curve	Bias^2	1.45	1.65	2.32	4.30	0.79
	ED	18.15	11.54	5.42	6.40	7.88
	RMSE	16.60	8.80	1.59	2.00	13.68
Flat response	Bias^2	1.14	0.82	0.28	0.22	1.51
	ED	18.07	8.08	0.77	0.68	7.74
	RMSE	17.32	14.13	11.24	15.29	20.34
Displacement	Bias^2	2.79	3.30	3.36	8.79	2.51
	ED	18.55	13.97	7.97	12.78	13.37
	RMSE	23.98	16.65	10.97	27.72	14.07
Sharp peak	Bias^2	16.01	11.63	6.81	24.45	0.86
	ED	19.09	11.05	3.86	10.85	7.94

Table 1: Estimated lag function RMSE, squared bias and effective degrees of freedom for five different types of lag curve shape, averaged across 200 simulated data sets. RMSE and Bias^2 have each been scaled by 10^3.

The models were assessed in terms of their ability to recover the true lag curve, as measured by root-mean squared error (RMSE) and squared bias (Bias^2) of the estimates, defined for the jth simulation as $\left(\frac{1}{51}\sum_{i=0}^{50}(\hat{\beta}_i^{(j)} - \beta_i^{(j)})^2\right)^{1/2}$ and $\left(\frac{1}{51}\sum_{i=0}^{50}\hat{\beta}_i^{(j)} - \beta_i^{(j)}\right)^2$, respectively. Following Hodges and Sargent [2001], model complexity was recorded for fitted model $\mathcal{M}_i^{(j)}$ using the effective degrees of freedom $\text{ED} = \text{tr}[\mathbf{X}(\mathbf{X}^\top\mathbf{X} + \hat{\mathbf{S}}_{\mathcal{M}_i^{(j)}})^{-1}\mathbf{X}^\top]$ where $\hat{\mathbf{S}}_{\mathcal{M}_i^{(j)}}$ is the estimated penalty component for the ith model fitted to the jth simulation. In particular, $\hat{\mathbf{S}}_{\mathcal{M}_1^{(j)}} = \hat{\lambda}^{(j)}\text{diag}(1, 2, \ldots, p+1)$; $\hat{\mathbf{S}}_{\mathcal{M}_2^{(j)}} = \hat{\lambda}^{(j)}\mathbf{P}$; $\hat{\mathbf{S}}_{\mathcal{M}_3^{(j)}} = \hat{\mathbf{Q}}^{(j)}$ and $\hat{\mathbf{S}}_{\mathcal{M}_4^{(j)}} = \hat{\lambda}^{(j)}\mathbf{P} + \hat{\rho}^{(j)}\mathbf{I}$. For models $\mathcal{M}_1$–$\mathcal{M}_4$ the model matrix $\mathbf{X}$ was identical, resulting from assuming $(2/3) \times (p+1) = 34$ uniformly spaced B-spline basis functions to represent the lag curve, such that the (i,k)th element, $\mathbf{X}_{i,k} = \sum_{j=0}^{p} x_{i-j}B_k(j)$. Since lag curve smoothness for $\mathcal{M}_5$ is decided by finding the number of logarithmically spaced knots that minimises AIC, the effective dimension in this case is the number of knots, $\text{ED} = n_z^{(j)}$ for the jth simulation.

Table 1 shows the average RMSE, Bias^2 and ED for each model and lag curve scenario combination. The adaptive model, $\mathcal{M}_3$, outperformed all of the models in terms of RMSE across all lag curve scenarios. Large separations were observed in model complexity across each scenario, with the adaptive smoothing $\mathcal{M}_3$ resulting in the lowest ED across all non-null scenarios, ranging between

3.86 to 7.97 compared to 6.4 and 19.09 for the other models. A particularly concerning result is that only $\mathcal{M}_3$ and $\mathcal{M}_4$ correctly result in degrees of freedom ≈ 1 under the FLAT RESPONSE scenario, compared to a range of 7.7 and 18 for the others, which suggests that other approaches may falsely identify structure when none exists at all. Bias was generally low across all models and scenarios, although higher values were noted across $\mathcal{M}_1$–$\mathcal{M}_4$ under the SHARP PEAK scenario. As might be expected, the log-spaced knots of $\mathcal{M}_5$ performed favourably when lag influence declined rapidly from 0 (SHARP PEAK), and much less so when the strongest lag effect was larger than 0 (DELAYED PEAK, DISPLACEMENT). The comparison clearly highlighted the need for care in choosing a lag smoother in practical application, and that the quality of fit strongly depends on the underlying lag function. Furthermore, the simulations provide encouraging evidence that the newly proposed adaptive model is far more robust to these underlying differences, and might be a suitable choice when little is known about the structure being estimated.

3.2 DLM misspecification with maximum lag p

Next, simulation was used to explore robustness of $\mathcal{M}_1$–$\mathcal{M}_5$ to different choices of maximum lag p. The series x_t and y_t were generated exactly as in Section 3.1, using only the DISPLACEMENT lag function with 50 lags. The models $\mathcal{M}_1$–$\mathcal{M}_5$ were fitted to the simulated x_t and y_t, assuming maximum lags of $p = 50, 75, 100, 125$. To ensure the fairest comparison, the range of flexibility for each model was restricted so that $0 < \text{ED} < 2p/3$, and also ensured that lag curve flexibility could grow proportionally with p. For each of 200 fits, the RMSE of the lag curve estimate, ED and squared bias were recorded and the results are summarised in Figure 2.

Figure 2 clearly illustrates, for $\mathcal{M}_1$ and $\mathcal{M}_5$, that ED increases steeply with p, even though the underlying true lag function is identical. Furthermore, both the RMSE and ED are substantially smaller for the adaptive model $\mathcal{M}_3$ compared to any of the others in the comparison, and indicate much better overall performance. The new adaptive approach $\mathcal{M}_3$ is more robust to user misspecification of 'large' p, and the simulations raise concerns about the potential for overfitting using existing approaches. However, it is noted that the effective degrees of freedom for $\mathcal{M}_3$ are not constant and do increase modestly with p, from 6.7 ($p = 50$) to 7.4 ($p = 125$). Model $\mathcal{M}_2$ resulted in a comparatively favourable balance between RMSE and degrees of freedom, reinforcing the suggestion that adaptive models are preferable choices.

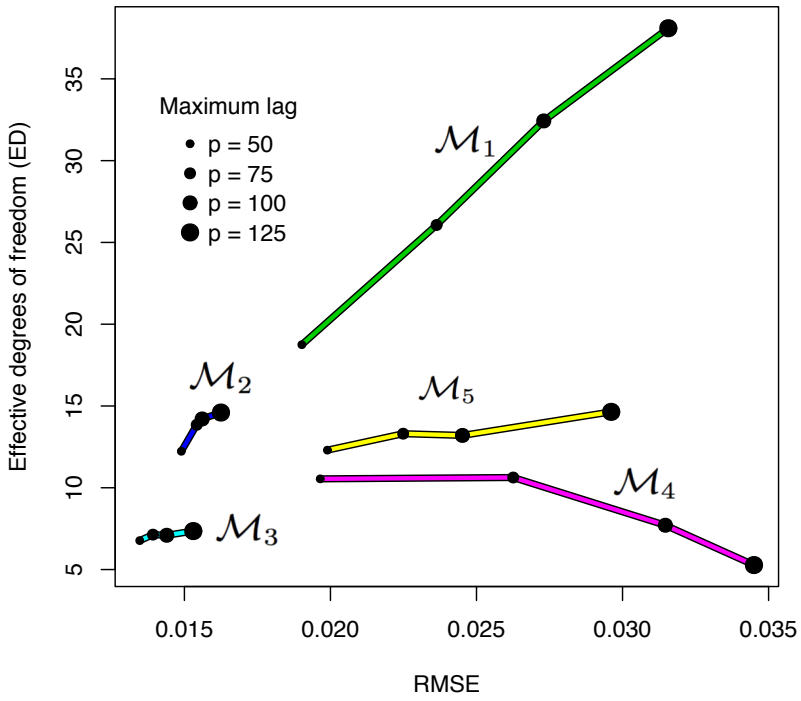


Figure 2: Effective number of parameters and RMSE for each of $\mathcal{M}_1$–$\mathcal{M}_5$. Increasing diameter of the plotting symbols indicates larger numbers of maximum lags assumed in advance of model fitting.

4 Illustrations with real data

Four applications illustrate how the adaptive lag curve behaves on noisy real data. Both use the accompanying software with its default settings: the random walk prior on the log precisions, an exact zero constraint at the maximum lag, $K = \lceil 2(p+1)/3 \rceil$ basis functions, and the collapsed NUTS sampler with two chains of 1000 retained draws. Effective degrees of freedom are reported for the lag curve alone, excluding unpenalised terms. The scripts that reproduce them are in the software repository.

4.1 Daily stock market data

Prices are not stationary, so they are transformed to percentage log returns $r_t = 100 \log(P_t/P_{t-1})$. Daily closing prices of the S&P 500 index and of Apple, and closing values of the VIX implied volatility index, were obtained for 2007 to 2016, 2,306 trading days spanning the 2008 crisis. Figure 3 shows four lag curves. Panels (a) and (b) regress the next day's return on the preceding 20 days'

returns, the linear information that a trading rule could exploit. For the index the curve is negative at lag one, −0.10 (95% interval −0.15 to −0.06) per percentage point, and returns to zero within a week: a short-lived reversal with cumulative effect −0.23 (−0.42 to −0.03) and 4.7 effective degrees of freedom. For Apple the curve is indistinguishable from zero at every lag, with 1.7 effective degrees of freedom; the adaptive smoother reports an absence of structure rather than inventing it. Panel (c) regresses the daily change in log VIX on the same day's and the preceding 60 days' index returns. The contemporaneous coefficient is −4.1 (−4.2 to −3.9), so implied volatility rises about 4 per cent for each percentage point the index falls; it is followed by a small positive rebound over the following week as the shock partly reverses, and by nothing beyond about ten days. Panel (d) regresses the log absolute return, a measure of realised volatility, on its own past over 100 days. The curve is strongest at lags two to three, decays over about a month and remains small but mostly positive thereafter, the long memory of volatility; the cumulative elasticity is 0.84 (0.75 to 0.94) and the smoother describes 100 lags with 7.9 effective degrees of freedom. In each panel the band widens where the data are least informative and the curve is pinned to zero at the maximum lag.

4.2 Intraday order flow and prices

At the other extreme of time scale, the sample data distributed with the QuantConnect Lean repository contain every trade and quote for the exchange traded fund SPY and for the stock BAC on the five trading days from 7 to 11 October 2013. They were reduced to one-second series: the midquote from the best bid and offer of the four most active venues, its return in basis points, the order-flow imbalance (the number of buyer-initiated minus seller-initiated trades, signed against the prevailing midquote), volume and the quoted spread. Lags were formed within each day, so that the first second of a day is never regressed on the previous afternoon, and B-splines of time of day entered as unpenalised terms to absorb the intraday pattern in volume and volatility. Each series has 117,000 seconds, and all fits took the default settings. Figure 4 shows the results.

Panel (a) regresses the midquote return on the order-flow imbalance of the same second and the 120 before it, and panel (b) shows the cumulative sum of these coefficients, the response of the price to one unit of order flow as a function of time since the trade, with its credible band. One more buy than sell trade moves the midquote by 0.0061 basis points in the same second; the response

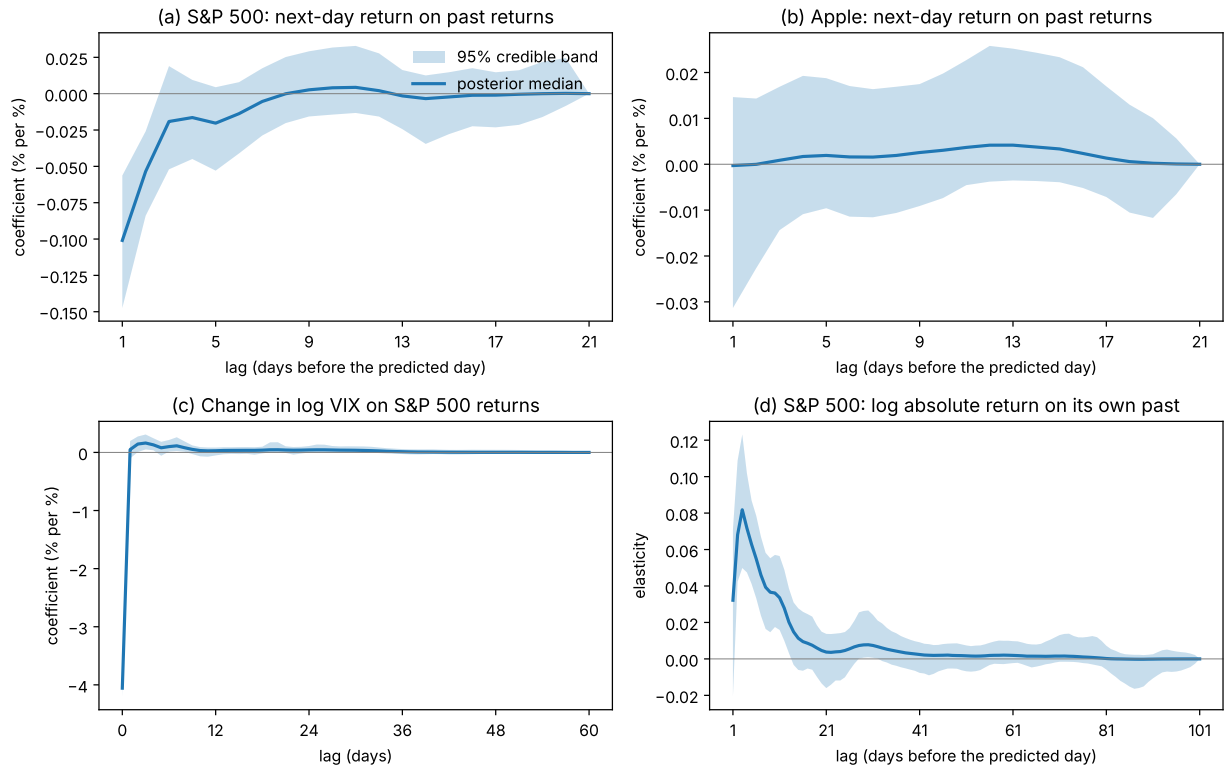


Figure 3: Lag curves for daily stock market data, 2007 to 2016: posterior medians with 95% credible bands. (a) Next-day S&P 500 return on the previous 20 days' returns; (b) the same for Apple; (c) daily change in log VIX on contemporaneous and lagged S&P 500 returns; (d) log absolute S&P 500 return on its own past over 100 days.

continues to build for about seven seconds, to 0.0093, and about 15 per cent of it then decays away over the following minute, leaving a permanent impact of 0.0079 (0.0074 to 0.0084). This is the shape that microstructure models describe with a transient propagator [Hasbrouck, 1991, Bouchaud et al., 2004], estimated here without assuming its functional form. Panels (c) and (f) regress the imbalance on its own past over 300 seconds for SPY and BAC. The coefficient at one second is 0.16 for SPY and 0.11 for BAC, but the tails carry most of the memory: the coefficients over five minutes sum to 0.38 and 0.57 respectively, and the curve for BAC, a large-tick stock whose queue dynamics encourage order splitting, remains positive to about 100 seconds [Lillo and Farmer, 2004]. Panels (d) and (e) show the two directions of the relation between volume and volatility: a thousand shares traded a second earlier raise the next second's absolute return by 0.001 basis points, with a decay over about a minute and a secondary bump at six to seven seconds that coincides with the time taken to absorb order flow in panel (b); and one basis point of absolute return attracts about 1,400

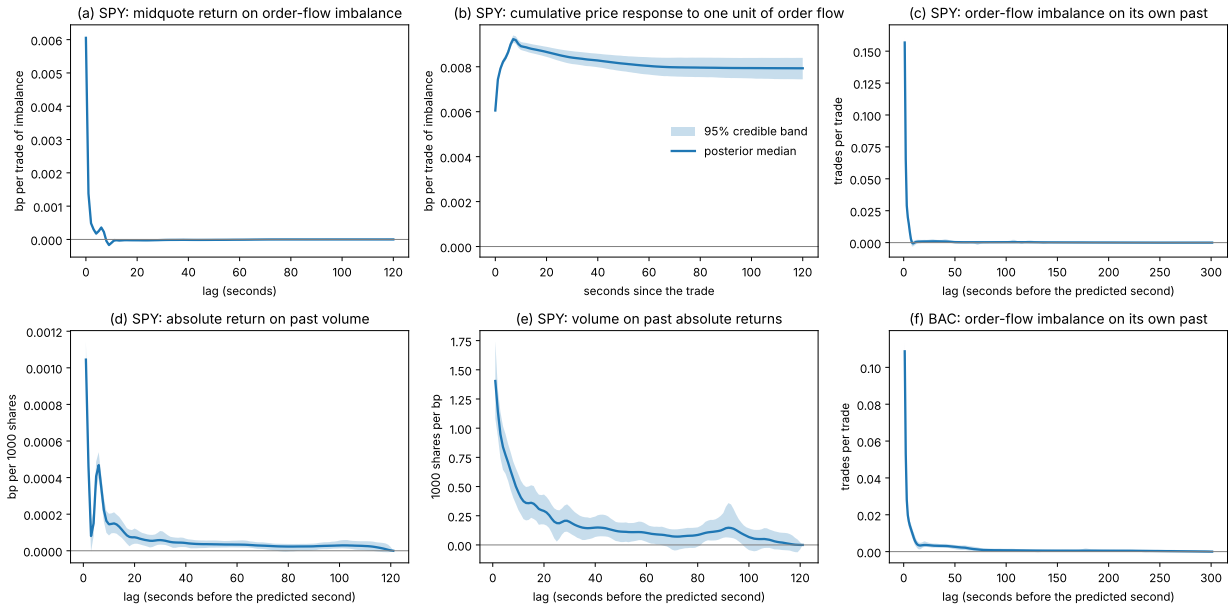


Figure 4: Intraday lag curves at one-second resolution, 7 to 11 October 2013. (a) SPY midquote return on order-flow imbalance; (b) the cumulative sum of (a), the price response to one unit of order flow; (c) SPY order-flow imbalance on its own past; (d) SPY absolute return on past volume; (e) SPY volume on past absolute returns; (f) BAC order-flow imbalance on its own past.

additional shares in the following second and some 22,000 over the following two minutes. The six curves use between 8 and 12 effective degrees of freedom for 120 to 300 lags.

4.3 Air pollution and mortality in Chicago

The setting that motivated much of the distributed lag literature is the delayed effect of air pollution on daily mortality, and in particular the question of mortality displacement raised in Section 1. The `chicagoNMMAPS` data distributed with the R package `dlnm` [Gasparrini, 2011] contain daily counts of all-cause, cardiovascular and respiratory deaths in Chicago from 1987 to 2000, 5,114 days with means of 115, 51 and 9 deaths per day, together with PM10 (mean 34 micrograms per cubic metre, 5 per cent of days interpolated), temperature and dew point. Log deaths were regressed on 40 lags of PM10 with the usual adjustments of this literature as unpenalised terms: a B-spline of time with seven degrees of freedom per year for trend and season, weekday indicators, and B-splines of temperature and dew point, with AR(1) errors. Coefficients are expressed as the percentage change in deaths per 10 micrograms per cubic metre.

Figure 5 shows the three lag curves and their cumulative sums. For all-cause deaths the curve

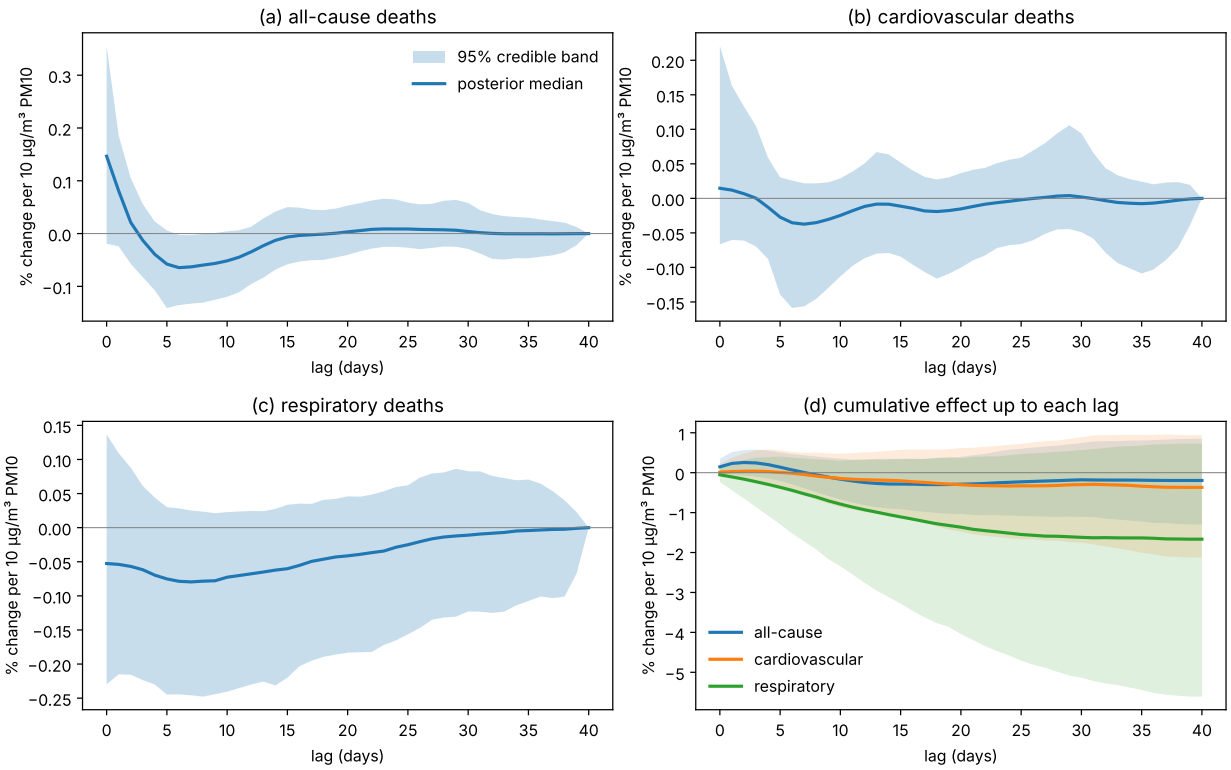


Figure 5: Chicago, 1987 to 2000: percentage change in daily deaths per 10 micrograms per cubic metre of PM10 at lags 0 to 40 for (a) all-cause, (b) cardiovascular and (c) respiratory deaths, with 95% bands, and (d) the cumulative effect up to each lag.

is positive on the day of exposure, 0.15 per cent (95% interval −0.02 to 0.36), falls through zero within three days and is negative from about lag four to lag ten, where the band excludes zero, before returning to zero. The cumulative effect over 40 days is −0.20 per cent (−1.30 to 0.85): deaths are brought forward by a few days rather than added, the displacement pattern of Schwartz [2000] and Zanobetti et al. [2000], and the lag curve exhibits it without any lag structure having been specified in advance. The curve uses about five effective degrees of freedom. Cardiovascular deaths show the same shape with smaller amplitude, and respiratory deaths, at nine per day, are too few to show anything beyond a wide band around zero.

4.4 COVID-19 cases and deaths

Deaths from COVID-19 follow infections after a delay of a few weeks, so the lag curve relating daily deaths to earlier cases estimates both the size of the relationship and the shape of the delay. Daily counts for the United States from June 2020 to March 2021, 304 days with a mean of 1,471 deaths

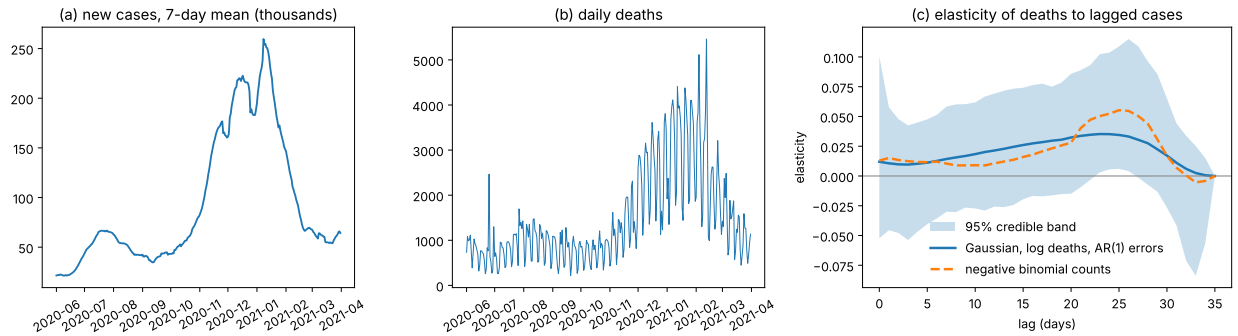


Figure 6: COVID-19 in the United States, June 2020 to March 2021. (a) Seven-day mean of new cases; (b) daily deaths; (c) elasticity of deaths to cases at lags 0 to 35 under the Gaussian model for log deaths with AR(1) errors (posterior median and 95% band) and under a negative binomial model for the counts (dashed).

per day, were taken from the compilation by The New York Times; the period precedes any effect of vaccination on the case fatality ratio. The exposure is the log of the seven-day trailing mean of new cases and the response is log deaths, so the lag coefficients are elasticities. Weekday indicators absorb the day-of-week pattern in death reporting and the errors follow an AR(1) process to allow for the remaining serial correlation. Figure 6 shows the data and the fitted curve with $p = 35$. The cumulative elasticity is 0.74 (0.66 to 0.81): a 10 per cent rise in cases is followed by roughly a 7 per cent rise in deaths, the shortfall from one reflecting the fall in the case fatality ratio over the period. The curve rises to a peak at about three weeks and returns to zero by five, using under three effective degrees of freedom, but the individual lag coefficients are poorly identified, because the smoothed case series is highly collinear across lags: the data determine the total and the timing, and the adaptive prior supplies the smoothness. Deaths reported on Tuesday to Friday are about 2.5 times those reported on a Sunday. A negative binomial model for the raw counts, shown dashed in Figure 6, agrees on the total elasticity, 0.73 (0.68 to 0.77), and on the peak, but its sampler did not converge satisfactorily, with many divergent transitions and $\hat{R}$ of 1.4: the dependence between the curve and its smoothing parameters that the Gaussian model integrates out remains present in a count model, and a collapsed or approximately collapsed scheme for count responses is a topic for further work.

5 Discussion

This paper makes several important contributions. It was shown by simulation in Section 3 that estimation of lag structure can strongly depend on the type of smoothing model that is assumed. Several existing DLM models were shown to be non-robust to the choice of maximum lag p, even when the underlying lag function is identical, which suggests that the interpretation of lag estimates should be made with caution. A new model has been proposed that combines automatic adaptive smoothing with a pragmatically large choice of p to ensure simple and flexible smoothing of the lag curve that avoids sensitivity to the choice of p. The real data illustrations in Section 4 show the adaptive curve reporting no structure where the data contain none, a sharp and short-lived response where there is one, a slowly decaying response over a hundred lags, the build-up and partial decay of price impact over seconds, and the displacement of deaths by a few days after a polluted day, in each case with far fewer effective parameters than the number of lags. The new approach provides users of DLMs with a new way to explore their data with confidence that the estimates are not contaminated by artefacts that result from particular model choices.

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